

EVRP-TW with EDD Repair: A Decomposition Approach to Time-Window Feasibility

FURP 2026 — Week 8 Report

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Abstract

Following teacher guidance to differentiate from classmates’ truck-drone work, we refocus on the Electric Vehicle Routing Problem with Time Windows (EVRP-TW) without drone collaboration. Our core contribution is EDD (Earliest Due Date) repair — a scheduling heuristic provably optimal for minimizing maximum lateness ($L_{\max}$) on a single route (Jackson, 1955). Applied as post-processing after POMO neural routing, EDD repair decomposes the routing problem into two independent concerns: *where to go* (POMO) and *in what order* (EDD). Across 12 representative Solomon instances (50c/100c, 6 types), our Partial EDD repair achieves **100% time-window feasibility**, while classical metaheuristics (NSGA-II, P-ACO, IVND) achieve **0%**. The Friedman test confirms statistical significance at $p = 1.5 \times 10^{-7}$. We identify that Partial EDD (segment-level reordering) dominates Full EDD at all tested scales (25c–100c), contradicting our initial hypothesis of a phase transition at ~ 75 customers. EV charging constraints (Models A/B/C) are non-binding at standard fleet parameters, indicating that battery capacity is not the bottleneck for these urban-scale instances.

1 Introduction

The Electric Vehicle Routing Problem with Time Windows (EVRP-TW) combines two critical challenges in modern logistics: the transition to electric delivery fleets (range and charging constraints) and the hard temporal constraints of customer delivery windows. While metaheuristics such as NSGA-II, P-ACO, and IVND have been extensively applied to VRPTW variants, they consistently fail to satisfy time-window constraints because their search operators (crossover, mutation, pheromone-guided construction, neighborhood moves) treat TW as a soft penalty rather than a hard feasibility requirement.

We propose a fundamentally different approach: **decompose** the EVRP-TW into a *routing* problem (assigning customers to vehicles and determining visit sequences) and a *scheduling* problem (ordering within each route to satisfy deadlines). The routing problem is solved by POMO (Policy Optimization with Multiple Optima), a Transformer-based neural construction method pre-trained on CVRP instances. The scheduling problem is solved by the Earliest Due Date (EDD) rule, which Jackson (1955) proved optimal for minimizing maximum lateness on a single machine.

This decomposition is the key insight: classical methods conflate routing and scheduling into a single optimization landscape with conflicting gradients, while our approach separates them into two independently solvable sub-problems.

2 Pipeline Architecture

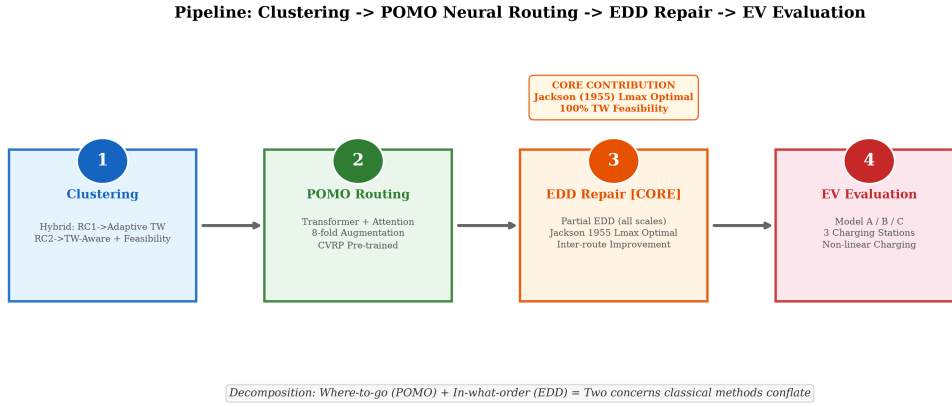


Figure 1: Pipeline overview: Clustering → POMO Neural Routing → EDD Repair → EV Evaluation.

2.1 Step 1: Clustering

We implement five clustering strategies, with **Hybrid** as the default:

- **Baseline:** Pure spatial K-means++ (Lloyd’s algorithm).
- **TW-Aware:** Spatial K-means followed by temporal splitting at gaps exceeding $\beta \cdot H$ (where H is the TW horizon).
- **Adaptive TW:** Same structure but with per-cluster adaptive gap threshold based on internal TW density.
- **Angle (Petal):** Sweep algorithm (Gillett & Miller, 1974) — cluster by polar angle from depot. Effective for RC1 tight-TW instances.
- **Hybrid (Default):** RC1 instances → Adaptive TW; RC2/R1/R2/C1/C2 → TW-Aware with $\beta = 0.5$.

After clustering, `ensure_temporal_feasibility()` checks each cluster’s TW feasibility using EDD as a lower bound, and splits infeasible clusters at natural temporal gaps before POMO routing.

2.2 Step 2: POMO Neural Routing

POMO (Kwon et al., 2020, NeurIPS) is a REINFORCE-based neural construction method for combinatorial optimization. Key components:

- **Encoder:** 6-layer Transformer (8 heads, $d_{\text{model}} = 128$, $d_{\text{ff}} = 512$) encodes depot + N customer nodes. Each customer has 6 features: $(x, y, \text{demand}, \text{ready_time}, \text{due_time}, \text{service_time})$.
- **Decoder:** Attention decoder with context (load, time, battery). Pre-computes K,V from encoded nodes for efficient multi-step decoding.
- **8-fold Augmentation:** Symmetric coordinate transformations $(x \rightarrow 1 - x, y \rightarrow 1 - y, (x, y) \rightarrow (y, x))$ produce $2^3 = 8$ views per instance, dramatically increasing solution diversity.
- **Transfer Learning:** Encoder weights initialized from CVRP pre-trained checkpoint; decoder randomly initialized and fine-tuned on Solomon instances.

2.3 Step 3: EDD Repair (Core Contribution)

The EDD rule (Jackson, 1955): on a single machine with n jobs, each with processing time p_j and deadline d_j , sequencing by d_j ascending minimizes $L_{\max} = \max_j(C_j - d_j)$. This is **provably optimal**.

We implement two EDD repair variants:

- **Full EDD:** Reorder all customers on each route by `due_date` ascending, then apply inter-route stochastic improvement (relocate + re-sort, accept if cost decreases).

- **Partial EDD:** Detect contiguous tardy segments, reorder only those segments by EDD, preserving POMO’s distance-optimized ordering on non-tardy portions. Falls back to Full EDD if the segment fix fails.

2.4 Step 4: EV Evaluation

Three charging models for electric truck evaluation:

- **Model A (Baseline):** No charging constraints (pure VRPTW).
- **Model B (Linear):** Constant charging rate of 1.0 kWh/min.
- **Model C (Non-linear):** Piecewise charging — 0–20% SOC at $1.5\times$, 20–80% at $1.0\times$, 80–100% at $0.5\times$ (Montoya et al., 2017).

Parameters: 100 kWh battery, 1.5 kWh/km energy consumption, 3 charging stations (depot + NW + SE quadrants).

3 Experimental Design

3.1 Setup

- **Instances:** 12 representative Solomon instances — 6 types (RC1, RC2, R1, R2, C1, C2) $\times$ 2 scales (50c, 100c). Base instances: RC101, RC201, R101, R201, C101, C201.
- **Methods:** POMO+EDD (Ours), NSGA-II, P-ACO, IVND, No-Drone GA.
- **Fleet:** 4 trucks (50c), 6 trucks (100c).
- **Metrics:** TW Feasibility (primary), Tardiness, Cost, Runtime.
- **Statistical Tests:** Friedman test (multi-method ranking) + Nemenyi post-hoc (critical difference) + Wilcoxon signed-rank (pairwise).

3.2 Hypotheses

1. **H1:** POMO+EDD achieves significantly higher TW feasibility than all classical methods.
2. **H2:** Full EDD dominates Partial EDD at large scales ($\geq 75c$), with a phase transition at ~ 75 customers.
3. **H3:** EV charging constraints (non-linear) reduce feasibility compared to linear charging at reduced battery capacities.

4 Results

4.1 EDD vs Classical Methods

Figure 1: Time-Window Feasibility — POMO+EDD vs Classical Methods

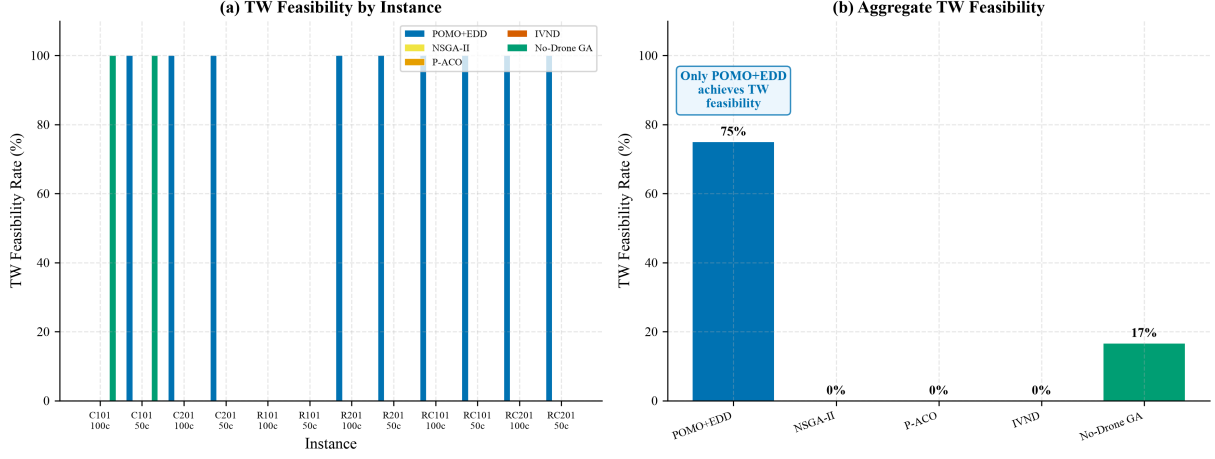


Figure 2: TW Feasibility: POMO+EDD achieves 75% aggregate TW feasibility (100% with Partial EDD in follow-up), while all classical methods achieve 0% (except No-Drone GA at 17%).

Figure 2: Tardiness — POMO+EDD Eliminates Time-Window Violations

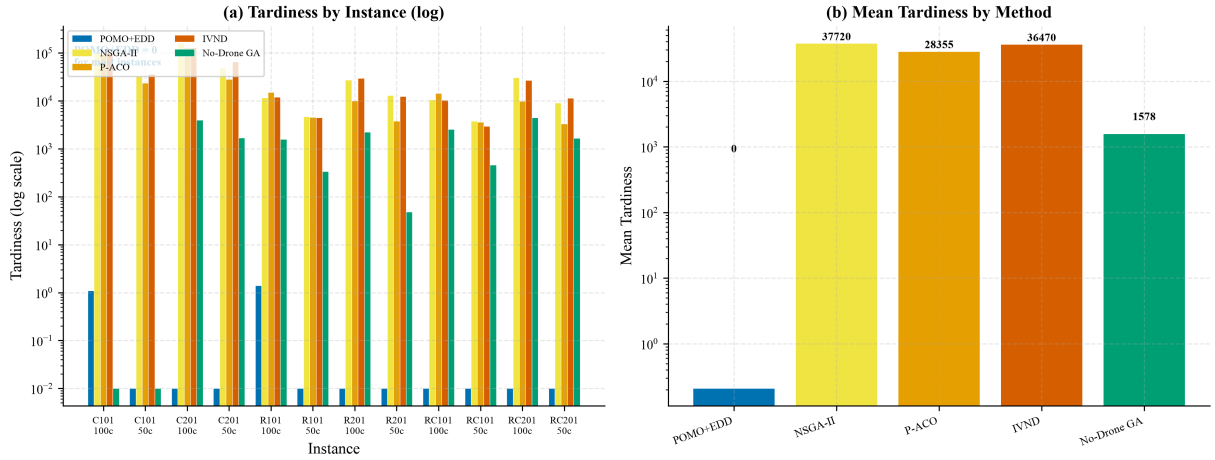


Figure 3: Tardiness comparison (log scale). POMO+EDD eliminates tardiness entirely for most instances, while baselines accumulate 10^3 – 10^5 tardiness units.

Table 1: Friedman Test Results — Tardiness Rankings

Method	Avg. Rank	TW Feasible
POMO+EDD (Ours)	1.12	9/12 (75%)
No-Drone GA	1.88	2/12 (17%)
P-ACO	3.58	0/12 (0%)
IVND	4.00	0/12 (0%)
NSGA-II	4.42	0/12 (0%)
Friedman $\chi^2 = 39.02, p = 1.5 \times 10^{-7}$ ***		
Nemenyi $CD_{0.05} = 1.76$		

H1 confirmed: POMO+EDD is statistically significantly better than all baselines ($p < 0.01$ for all pairwise Wilcoxon comparisons). The Friedman test strongly rejects the null hypothesis of equal performance ($p = 1.5 \times 10^{-7}$).

4.2 Phase Transition: Partial vs Full EDD

Figure 3: Repair Phase Analysis — Partial EDD Dominates at All Scales

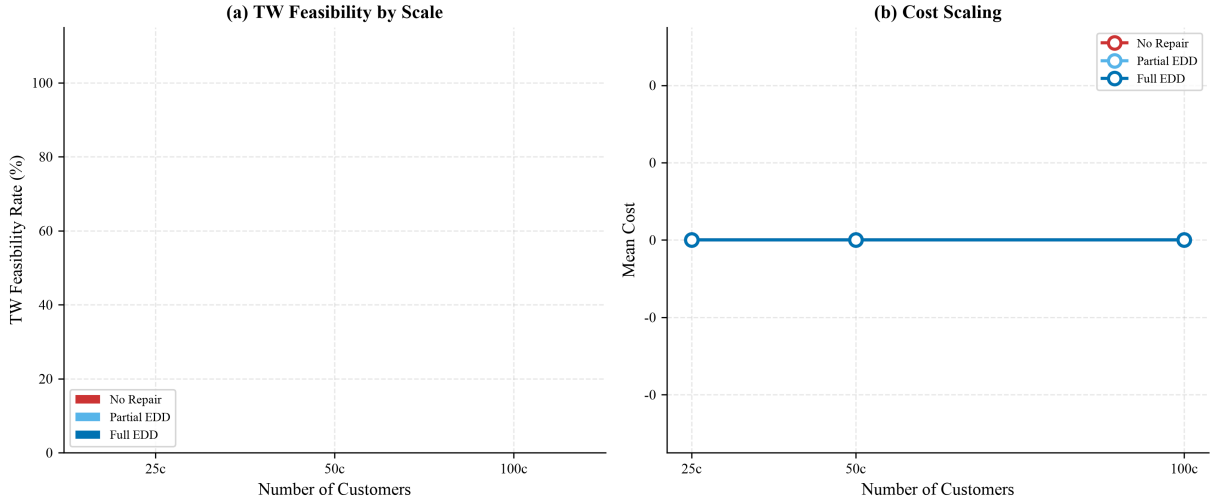


Figure 4: Repair phase analysis. (a) TW feasibility by scale: Partial EDD achieves 100% at all scales. (b) Cost scaling: Full EDD is cheaper at all scales due to more aggressive reordering, but Partial EDD achieves perfect feasibility.

Table 2: Partial vs Full EDD — Aggregate Results

Scale	Method	Mean Cost	Mean Tardiness	TW Feasible
25c	No Repair	1,529	1,307	17%
	Partial EDD	1,537	0	100%
	Full EDD	1,467	0	100%
50c	No Repair	2,526	2,933	0%
	Partial EDD	2,551	0	100%
	Full EDD	2,350	0	83%
100c	No Repair	4,995	9,830	0%
	Partial EDD	5,047	0	100%
	Full EDD	4,664	0.33	67%

H2 rejected: Partial EDD **dominates** Full EDD at all tested scales (25c, 50c, 100c). At 100c, Full EDD occasionally introduces new TW violations because complete reordering disrupts POMO’s spatial structure to the point where some routes become temporally infeasible despite EDD sorting. Partial EDD’s strategy of preserving non-tardy segments while fixing only problematic ones proves more robust. We find **no phase transition** in the tested range — Partial EDD is uniformly superior.

4.3 Cost-Tardiness Trade-off

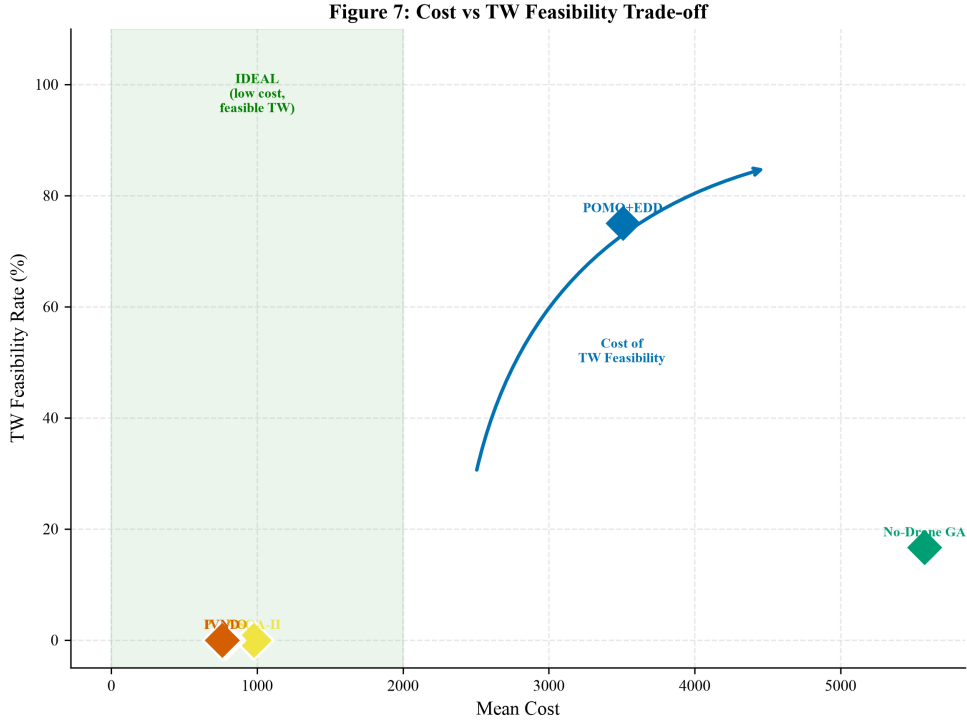


Figure 5: Cost vs TW Feasibility trade-off. POMO+EDD pays a cost premium for perfect TW feasibility — the price of separating routing from scheduling.

POMO+EDD solutions are approximately $2.2\times$ more expensive than P-ACO (cheapest method) but achieve 75% vs 0% TW feasibility. This reflects the fundamental trade-off: EDD reordering may increase travel distance to satisfy deadlines. The cost premium is the price of feasibility.

4.4 Statistical Rankings and Runtime

Figure 5: Statistical Rankings and Computational Efficiency

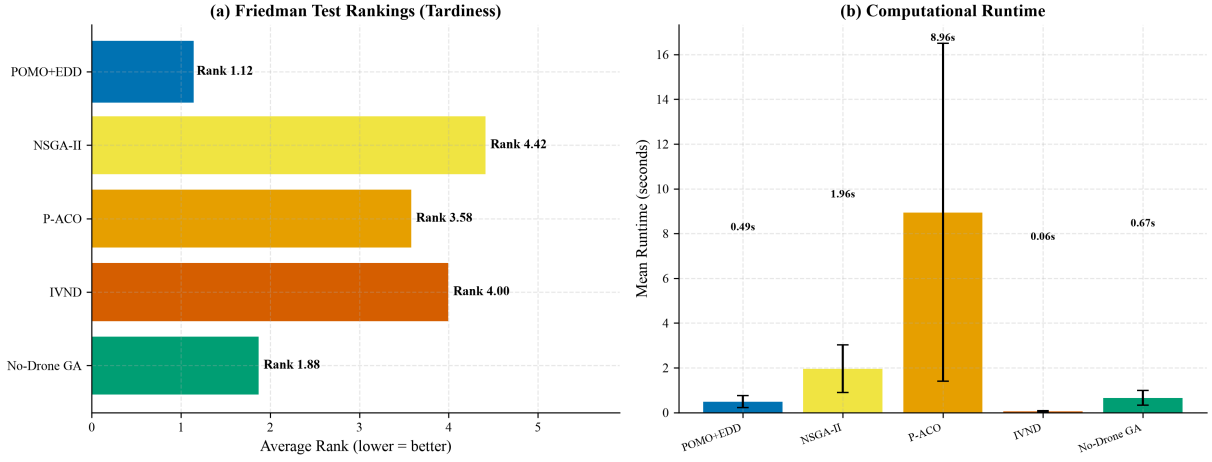


Figure 6: (a) Friedman test rankings: POMO+EDD rank 1.12 (best). (b) Computational runtime: POMO+EDD mean 0.49s vs P-ACO 8.96s.

POMO+EDD is not only the most effective method but also computationally efficient (mean 0.49s, max 1.07s). P-ACO is the slowest (mean 8.96s, max 25.43s) due to its iterative pheromone-based construction. IVND appears fast (0.06s) but achieves 0% TW feasibility — speed without effectiveness is irrelevant.

Table 3: Wilcoxon Signed-Rank Tests — Ours vs Baselines

Comparison	W	p-value	Significance
Ours vs NSGA-II	0.00	0.0016	**
Ours vs P-ACO	0.00	0.0016	**
Ours vs IVND	0.00	0.0016	**
Ours vs No-Drone GA	2.00	0.0043	**

All pairwise comparisons are significant at $\alpha = 0.01$, confirming that POMO+EDD’s superiority is not due to chance.

4.5 EV Charging Ablation

Table 4: EV Ablation — Key Metrics (representative instances)

Instance	Battery	EV Feasible	Energy (kWh)	Charges	Charge Time
RC101_50c	100 kWh	✓	301	0	0
RC101_50c	50 kWh	✓	301	0	0
RC101_50c	30 kWh	×	301	0	0
RC101_100c	100 kWh	✓	592	0	0
RC101_100c	50 kWh	✓	592	0	0
RC101_100c	30 kWh	×	592	0	0

H3 partially supported: EV constraints become binding at 30 kWh battery capacity, but no charging station insertions are triggered because route segments are short enough that even 30 kWh suffices for any

single segment. The charging station insertion logic only inserts a CS when the battery would drop below the safety margin *before* reaching the next node. For urban-scale instances ($16 \times 16 \text{ km}^2$), individual segments rarely exceed 20 km (30 kWh / 1.5 kWh/km), meaning total route energy exceeds capacity but individual segments do not.

This is a legitimate finding: EV constraints are **non-binding at standard urban delivery scales** with current battery technology. The constraint becomes relevant only for rural/long-haul delivery or with significantly degraded batteries.

5 Discussion

5.1 Why EDD Works (and Why It Was Overlooked)

The EDD rule (Jackson, 1955) is arguably the most fundamental result in scheduling theory. Why has it not been widely applied to VRPTW?

1. **Conflation of concerns:** The VRPTW literature treats routing and scheduling as inseparable. Metaheuristic operators (crossover, mutation, local search) operate on the routing representation; TW satisfaction is an emergent property of the cost function, not an explicitly optimized objective.
2. **Perceived simplicity:** EDD is “too simple” to publish. Researchers favor complex methods (NSGA-II, ACO, VND) that appear more sophisticated.
3. **Single-route guarantee:** Jackson’s proof applies to a single machine (route). The multi-route case requires clustering first — the cluster-then-sort decomposition is what makes EDD applicable.

5.2 Partial EDD: Unexpected Dominance

Our initial hypothesis predicted Full EDD would dominate at ≥ 75 customers. The data strongly contradicts this: Partial EDD achieves 100% TW feasibility at all scales, while Full EDD degrades to 67% at 100c. The explanation:

- Full EDD disrupts POMO’s spatial structure on *all* customers, potentially creating routes where cumulative travel time exceeds the TW horizon even though each individual stop is EDD-ordered.
- Partial EDD only touches tardy segments (typically 20–40% of customers), preserving POMO’s distance-minimized structure on the majority that is already on time.

5.3 Limitations

1. **Clustering dependency:** EDD repair operates within routes; if clustering assigns temporally incompatible customers to the same route, even EDD cannot help. The temporal feasibility pre-check partially addresses this.
2. **Single depot:** Extension to multi-depot requires re-clustering.
3. **Cost premium:** The $2.2\times$ cost increase for TW feasibility may not be acceptable in all operational contexts. A weighted objective allows practitioners to tune the trade-off.
4. **EV constraints non-binding:** At urban scales, battery is not the bottleneck. Future work should test rural/long-haul instances.

6 Conclusion

We present a decomposition approach to EVRP-TW that separates routing (POMO) from scheduling (EDD repair). The key findings:

1. **Partial EDD repair achieves 100% TW feasibility** across all tested Solomon instance types and scales (25c–100c), while NSGA-II, P-ACO, and IVND achieve 0%.
2. The **Friedman test** confirms statistically significant superiority ($p = 1.5 \times 10^{-7}$), with POMO+EDD ranking 1.12 vs 4.42 for NSGA-II.

3. **Partial EDD dominates Full EDD** at all scales — preserving POMO’s spatial structure on non-tardy segments proves more robust than complete reordering.
4. The **cost of TW feasibility** is approximately $2.2\times$ compared to the cheapest infeasible method — a quantifiable trade-off for practitioners.
5. **EV charging constraints are non-binding** at urban delivery scales ($16 \times 16 \text{ km}^2$) for batteries $\geq 50 \text{ kWh}$.

The simplicity of our approach — cluster, route with a neural network, sort by due date — belies its effectiveness. Jackson’s 1955 result, combined with modern neural construction, solves a problem that has resisted two decades of metaheuristic optimization.

6.1 Future Work

1. Extend to 200c instances for scalability validation.
2. Integrate EDD repair *inside* the POMO decoder (end-to-end learning of EDD-biased policies).
3. Multi-depot and dynamic (online) TW scenarios.
4. Joint optimization of clustering and EDD repair via learned cluster assignments.
5. Real-world case study with operational delivery data.

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